

Cycle	state	PC	MAR	MBR	IR	ACC	Bus driver	Active loads	Notes
1	fetch 1	0x000	-	-	-	0x0000	bus-pc	mar_ld	MAR ← 0x000
2	fetch 2	0x000	0x000	-	-	0x0000	bus-mem	mbr_ld, pc-inc	MBR ← mem[0x000] PC ← 0x001
3	fetch 3	0x001	0x000	0x200A	-	0x0000	bus-nbr	ir_ld	IR ← 0x200A
4	load	0x001	0x000	0x200A	0x200A	0x0000	bus-ir-se	acc_ld	ACC ← sign-xt(0x00A) = 0x000A
5	fetch 1	0x001	0x000	0x200A	0x200A	0x000A	bus-pc	mar_ld	MAR ← 0x001
6	fetch 2	0x001	0x001	0x200A	0x200A	0x000A	bus-mem	mbr_ld	MBR ← mem[0x001] PC ← 0x002
7	fetch 3	0x002	0x001	0x4100	0x200A	0x000A	bus-mbr	ir_ld	IR ← 0x4100
8	direct 1	0x002	0x001	0x4100	0x4100	0x000A	bus-ir-lower	mar_ld	MAR ← 0x100
9	direct 2	0x002	0x100	0x4100	0x4100	0x000A	bus-mem	mbr_ld	MBR ← 0x0000
10	store 1	0x002	0x100	0x0000	0x4100	0x000A	bus-acc	mbr_ld, mem-we	ACC → bus MBR, mem[0x100]
11	fetch 1	0x002	0x100	0x000A	0x4100	0x000A	bus-pc	mar_ld	MAR ← 0x002
12	fetch 2	0x002	0x002	0x000A	0x4100	0x000A	bus-mem	mbr_ld, pc-inc	MBR ← mem[0x002]
13	fetch 3	0x003	0x002	0x3100	0x4100	0x000A	bus-mbr	ir_ld	IR ← 0x3100
14	direct 1	0x003	0x002	0x3100	0x3100	0x000A	bus-ir-lower	mar_ld	MAR ← 0x100
15	direct 2	0x003	0x100	0x3100	0x3100	0x000A	bus-mem	mbr_ld	MBR ← 0x000A
16	load 1	0x003	0x100	0x000A	0x3100	0x000A	bus-nbr	acc_ld	ACC ← 0x000A
17	fetch 1	0x003	0x100	0x000A	0x3100	0x000A	bus-pc	mar_ld	MAR ← 0x003
18	fetch 2	0x003	0x003	0x000A	0x3100	0x000A	bus-mem	mbr_ld, pc-inc	MBR ← 0x700F PC ← 004
19	fetch 3	0x004	0x003	0x7101	0x3100	0x000A	bus-mbr	ir_ld	IR ← 0x7101
20	direct 1	0x004	0x003	0x7101	0x7101	0x000A	bus-ir-lower	mar_ld	MAR ← 0x101
21	direct 2	0x004	0x101	0x7101	0x7101	0x000A	bus-mem	mbr_ld	MBR ← 0x0005
22	add 1	0x004	0x101	0x0005	0x7101	0x000A	bus-alu	acc_ld	ACC ← 0x00F
23	fetch 1	0x004	0x101	0x0005	0x7101	0x000F	bus-pc	mar_ld	MAR ← 0x004
24	fetch 2	0x004	0x004	0x0005	0x7101	0x000F	bus-mem	mbr_ld	MBR ← 0x0000
25	fetch 3	0x005	0x004	0x0000	0x7101	0x000F	bus-mbr	ir_ld	IR ← 0x0000
26	halt 1	0x005	0x004	0x0000	0x0000	0x000F	-	done	done = '1'